

Export Industry ESG Assessment & BRSR Reporting Framework

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Industry Profile:

Ecosystem Architecture and Value Chain

India's export industry does not operate as a single entity but rather as a highly integrated, complex value chain. The architecture of this ecosystem directly links farms, manufacturing factories, service providers, ports, freight forwarders, customs teams, and overseas buyers.

Because the ecosystem operates on such a massive scale across numerous sites and suppliers, even minor fluctuations in operational performance can have a material impact on lead times, costs, and reputational exposure. Consequently, an accurate industry profile must look beyond the immediate factory gate to encompass logistics corridors, outsourced operations, and buyer-mandated requirements. From an operational standpoint, this means that issues like sustainability or quality do not remain isolated within one department; they simultaneously affect procurement, operations, logistics, finance, and overall buyer confidence.

Economic Scale and Macro-Trends

The sector operates on a massive economic scale. Official government trade reporting indicates that merchandise exports reached US\$ 437.07 billion during the 2023-24 period, accompanied by services exports totaling US\$ 341.06 billion. The latest government estimates for April-February 2025-26 place total exports at an impressive US\$ 790.86 billion.

Analyzing historical trade trends, merchandise exports have seen substantial growth, rising from US\$ 310.34 billion in 2014-15 to the US\$ 437.07 billion mark in 2023-24, while imports for the same year stood at US\$ 678.21 billion. It is crucial to view this sector as inherently cyclical, driven by external variables such as global demand, fuel prices, and shifting freight conditions. Because of this, industry performance is best measured through metrics of operational intensity, resilience, and value addition rather than solely relying on absolute growth figures.

Product Clusters and Material Hotspots

The composition of export earnings is subject to rapid changes depending on commodity cycles, trade policies, and shipping variables. Currently, export growth is largely driven by key product categories including electronics, engineering goods, pharmaceuticals, rice, marine products, and gems and jewellery.

Each of these product clusters carries distinct ESG and operational "hotspots" that dictate their risk profile:

- **Textiles and Apparel:** High exposure to water usage, chemical handling, and labour-intensive processes. The operational focus here is on dyeing controls, effluent treatment, traceability, and worker welfare.

- **Engineering Goods:** Operations are heavy in energy use, scrap generation, and machining waste. Reporting and controls emphasize energy efficiency, metal recovery, and machine guarding.
- **Electronics:** The primary risks revolve around hazardous e-waste, data security, and supplier traceability. This requires robust cyber controls and comprehensive supplier mapping.
- **Food and Agro Exports:** Vulnerabilities center around water consumption, cold chain integrity, hygiene, and loss control, placing a premium on food safety and recall readiness.
- **Gems, Jewellery, and Leather:** Stakeholders heavily scrutinize product origin, chemical use, and ethics, necessitating stringent due diligence, responsible sourcing, and proof of product authenticity.

Regulatory Integration (The BRSR Framework)

The profile of the export industry is heavily shaped by the Business Responsibility and Sustainability Reporting (BRSR) framework, particularly for SEBI-listed entities. This framework mandates reporting across nine core principles. Furthermore, "BRSR Core" focuses attention on highly measurable indicators such as greenhouse gas footprints, water usage, energy consumption, waste management, and employee safety.

This framework maps directly to the realities of the export industry:

- **Principle 2 (Sustainable Goods):** Focuses on eco-design, safe products, and sustainable sourcing.
- **Principle 3 (Employee Wellbeing):** Covers safety, insurance, fair wages, and harassment prevention across the value chain.
- **Principle 5 (Human Rights):** Mandates grievance systems, due diligence, and remedy protocols.
- **Principle 9 (Customer Value):** Demands truthful labelling, rigorous quality control, and data protection.

Stakeholder Ecosystem and Materiality

A definitive aspect of the export industry is its broad and demanding stakeholder base. The most critical stakeholders include buyers, workers, contract labour, suppliers, local communities, lenders, logistics partners, and regulators. Across these groups, expectations converge heavily around product quality, delivery reliability, responsible sourcing, labour conditions, environmental controls, and strict compliance discipline.

The specific expectations of these groups shape the industry's day-to-day operations:

- **Buyers:** Demand rigorous traceability, audits, and ethical sourcing evidence through scorecards and corrective action plans.
- **Workers:** Expect safe working environments, fair pay, skill development, and accessible grievance channels.
- **Suppliers:** Require predictable order flows and compliance support, managed through supplier codes and onboarding audits.
- **Communities:** Look for localized job creation, transport safety, and effective pollution mitigation.
- **Regulators and Lenders:** Mandate absolute legal compliance, operational transparency, and resilience controls backed by assurance and disclosures.

ESG Framework:

The framework is applied through two primary structural components: the overarching principles and the measurable "Core" indicators.

1. The Nine BRSR Principles

The qualitative foundation of the BRSR framework is structured around nine principles that cover the full spectrum of Environmental, Social, and Governance (ESG) responsibilities. For an export business, these principles map out to the following operational areas:

- **Principle 1 (Ethics, Transparency & Accountability):** Focuses on board conduct, ethics training, and tracking fines or penalties.
- **Principle 2 (Sustainable Goods & Services):** Emphasizes eco-design, product safety, Extended Producer Responsibility (EPR), and sustainable sourcing.
- **Principle 3 (Employee Well-being & Value Chain Labour):** Covers workplace safety, insurance access, fair wages, and harassment prevention.
- **Principle 4 (Stakeholder Responsiveness):** Requires the structured engagement and mapping of material stakeholders.
- **Principle 5 (Human Rights):** Mandates human rights due diligence, grievance systems, and remedy protocols.
- **Principle 6 (Environment Protection & Restoration):** Covers the tracking of energy, water, waste, emissions, and Environmental Management Systems (EMS).
- **Principle 7 (Responsible Policy Engagement):** Involves transparent disclosure of trade association memberships and public advocacy positions.
- **Principle 8 (Inclusive Growth & Equitable Development):** Focuses on community impact, local development programs, and Social Impact Assessments (SIA).
- **Principle 9 (Customer Value & Responsibility):** Demands truthful product labelling, rigorous safety and quality controls, and data protection.

2. "BRSR Core" (Measurable Indicators)

While the nine principles guide the narrative, the **BRSR Core** narrows the spotlight to strict, quantifiable indicators. SEBI's guidance expects these disclosures to be comparable, strictly evidence-based, and ultimately useful to investors and stakeholders.

Under BRSR Core, export companies are asked to quantify hard metrics, such as:

- Greenhouse gas (GHG) footprint.
- Water footprint and energy footprint.
- Waste generation and recovery rates.
- Employee wellbeing and safety data (e.g., Lost Time Injury Frequency Rate or LTIFR).

Evidence and Boundaries

A critical part of applying this framework involves clearly stating the **reporting boundary**—whether the report covers a stand-alone entity or consolidated operations, and how it treats data from the broader value-chain. Because a disclosure is only as strong as its evidence trail, the framework requires linking these principles to specific, auditable sources like utility bills, HR logs, and board minutes.

Indicator Definition:

Environmental Indicators

- **Energy & Emissions:** Tracking energy intensity per unit output, the percentage of renewable electricity used, and Scope 1 and 2 greenhouse gas emissions intensity (measured in CO₂e per unit output).
- **Water Stewardship:** Measuring total water consumption, water intensity per unit output, and wastewater discharge categorized by treatment level.
- **Waste & Circularity:** Monitoring the percentage of waste recovered, and maintaining separate accounting for hazardous and non-hazardous waste streams like scrap, sludge, packaging, and e-waste.
- **Land & Biodiversity:** Assessing habitat disturbance, land conversion, and storm-water handling, especially during facility expansions.

Social Indicators

- **Workforce & Diversity:** Tracking total headcount across permanent, fixed-term, and contract staff, alongside metrics for female representation and pay equity.
- **Health & Safety:** Monitoring the Lost Time Injury Frequency Rate (LTIFR), total serious incidents, and direct spending on employee wellbeing.
- **Training & Skills:** Measuring training coverage as a percentage, total training hours, and competency completion rates.
- **Human Rights & Grievances:** Tracking the number of grievance cases, closure times, and indicators related to forced labour, child labour, or discrimination within the supply chain.
- **Customer & Product Responsibility:** Evaluating product conformity, labelling accuracy, complaint handling, and product recall readiness.

Governance Indicators

- **Board Oversight & Ethics:** Measuring the meeting cadence of ESG committees, completion of ethics training, and the resolution of compliance audits.
- **Compliance Control:** Maintaining logs for environmental permits, labour compliance, customs, and tax obligations to ensure no filings are missed.
- **Risk & Data Management:** Tracking residual risk levels, cyber security incidents, and the management of sensitive customer data.

- Policy Engagement: Disclosing affiliations with trade associations, chambers of commerce, and official public policy positions.

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Data Collection Methodology:

- **External Context Data:** The broader trade, labor, and demographic context is gathered from official Indian sources. Specific sources include Ministry of Commerce trade releases, MOSPI labor and population statistics, and SEBI's BRSR format and BRSR Core circular.
- **Internal Operational Evidence:** To remain audit-ready for limited assurance, companies must retain primary source evidence rather than relying solely on narrative summaries. This evidence includes meter logs, utility invoices, site registers, training sheets, grievance logs, and supplier questionnaires.
- **System Integration:** Data collection is integrated into single dashboards to help reconcile bills, meter data, production volumes, and improvement initiatives. Supplier audits are also used to collect data on labor standards, environment controls, and origin documentation.
- **Governance and Accountability:** Every data point must be connected to a measurable owner, a specific data source, and a management review cycle.

Table 1: Export product clusters and ESG hot spots

Export cluster	Likely ESG hotspot	Reporting emphasis
Textiles and apparel	Water, chemicals, labour and energy	Dyeing controls, effluent, traceability, worker welfare
Engineering goods	Energy, scrap, machining waste, safety	Energy efficiency, metal recovery, machine guarding
Electronics	Hazardous waste, data, supplier traceability	E-waste handling, cyber controls, supplier mapping
Food and agro exports	Water, cold chain, hygiene, loss control	Water reuse, food safety, storage and recall readiness
Gems / jewellery / leather	Origin, ethics, chemical use, labour	Responsible sourcing, due diligence and product authenticity

Table 2: BRSR principle cross-walk for export-industry reporting

Principle	Theme	Export relevance
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P1	Ethics, transparency and accountability	Board conduct, training and fines / penalties
P2	Sustainable goods and services	Eco-design, safe products, EPR and sustainable sourcing
P3	Employee well-being and value chain labour	Safety, insurance, access, wages and harassment prevention
P4	Stakeholder responsiveness	Material stakeholder mapping and structured engagement
P5	Human rights	Grievance systems, due diligence and remedy
P6	Environment protection and restoration	Energy, water, waste, emissions and EMS
P7	Responsible policy engagement	Trade association memberships and advocacy disclosure
P8	Inclusive growth and equitable development	Community impact, local development and SIA
P9	Customer value and responsibility	Truthful labelling, quality, safety and data protection

Table 3: Stakeholder materiality summary

Stakeholder group	Main expectation	Report response
Buyers	Traceability, quality, delivery, ethical sourcing	Buyer scorecards, audits and corrective action plans
Workers	Safe work, fair pay, skill growth, grievance access	Training, welfare, LTIFR and complaint closure
Suppliers	Predictable orders, fair terms, compliance support	Supplier codes, onboarding and audits
Communities	Jobs, transport safety, water, pollution control	Community meetings and local programs

Regulators / lenders	Legal compliance, controls, resilience, transparency	Policy logs, permits, assurance and disclosures
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Table 4: Social performance summary

Social area	Representative metric	Why it matters
Headcount	Permanent, fixed-term and contractor mix	Workforce structure should be transparent
Gender and inclusion	Female participation, pay equity, access	Diversity requires targeted support
Training and skills	Hours, coverage, competency completion	Productivity and safety depend on training
Safety and wellbeing	LTIFR, incidents, wellness spend	Prevention should be visible in the data
Grievance and remedy	Cases, closure time, recurrence	Closed-loop complaint handling is essential

Table 5: Social KPI dashboard

Metric	Unit	Baseline	Target	Owner
Total headcount	Persons	Baseline set	Annual target set	HR head
Female representation	%	Baseline set	Annual target set	HR / plant lead
Training coverage	%	Baseline set	Annual target set	L&D lead
LTIFR	Per million hours	Baseline set	Annual target set	EHS lead
Grievance closure	% within target time	Baseline set	Annual target set	HR / compliance

Table 6: Environmental KPI dashboard

Metric	Unit	Baseline	Target	Owner
Renewable electricity	% of total power	Baseline set	Annual target set	Utilities head
Energy intensity	Per unit output	Baseline set	Annual target set	Operations head
Water intensity	Per unit output	Baseline set	Annual target set	EHS / plant head
Waste recovery	% of waste recovered	Baseline set	Annual target set	EHS lead
Scope 1+2 intensity	CO2e per unit output	Baseline set	Annual target set	Sustainability lead

Table 7: Disclosure matrix appendix

BRSR principle / section	Key disclosure example	Evidence source	Frequency
P1 ethics	Training, fines, board oversight	Board minutes, LMS records	Annual / quarterly
P3 employee well-being	Welfare, incidents, LTIFR	EHS logs, HR files	Monthly / annual
P4 stakeholder responsiveness	Engagement logs and issues	Stakeholder register	Quarterly
P5 human rights	Risk review and grievance outcomes	Policy, complaints log	Quarterly
P6 environment	Energy, water, waste, emissions	Utility bills, meter logs	Monthly / annual
P7 policy advocacy	Memberships and positions	Association records	Annual
P8 inclusive growth	Community and SIA disclosures	CSR / community files	Annual

P9 customers	Quality, complaints, data privacy	QA / CRM / IT records	Monthly / annual
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1. Energy Consumption

- **Textile/Apparel Manufacturing:** A standard Indian garment production facility consumes an average of **30,000 to 40,000 kWh per day** (roughly **900,000 to 1.2 million kWh/month**).
- **Intensity Benchmark:** The specific energy consumption averages **2.58 kWh per kg** of dyed fabric, or roughly **12 Megajoules (MJ) per finished garment**.
- **Control Measure:** Top-tier factories are saving up to 25% of this energy by switching to variable speed drives (VSD) and high-efficiency electric steam generators.

2. Water Usage

Water is the most critical ESG metric for the textile and agro-export industries.

- **Current Indian Average:** Indian textile wet-processing (scouring, dyeing, and finishing) typically consumes **200 to 250 Liters of water per kg (L/kg)** of cotton fabric.
- **Daily Volume:** For a mid-sized export factory processing 5,000 kg of fabric daily, this translates to **1,000,000 to 1,250,000 Liters per day (L/day)**.
- **Target Benchmark:** Best-practice export facilities are implementing water recovery systems to bring this down to **<100 L/kg**.

3. Emissions & Pollution Control Measures

- **Carbon Footprint:** In the apparel sector, the average carbon emission is roughly **0.94 kg of CO2 per finished piece**, with steam generation accounting for up to 97% of the process-level carbon footprint. Heavy engineering exports (like steel) average **1.92 tonnes of CO2 per tonne** of product.
- **Pollution Controls:** The primary pollution control benchmark for Indian export facilities is achieving **Zero Liquid Discharge (ZLD)**. This ensures that the highly chemical, high-salinity wastewater generated during processing is 100% treated and recycled internally, rather than discharged into local groundwater.

4. Waste Quantity

- **Baseline Generation:** Waste generation depends entirely on the sector. In apparel, fabric fallout (scrap) can account for 10-15% of total raw material weight.

- **ESG Target:** Rather than raw kg/day, leading Indian manufacturers track the *recovery rate*. Top ESG performers aim to send **55% of waste to recycling** and **38% to co-processing** (using waste as fuel in cement kilns), driving total waste to landfill down to single digits.

5. Workforce Data

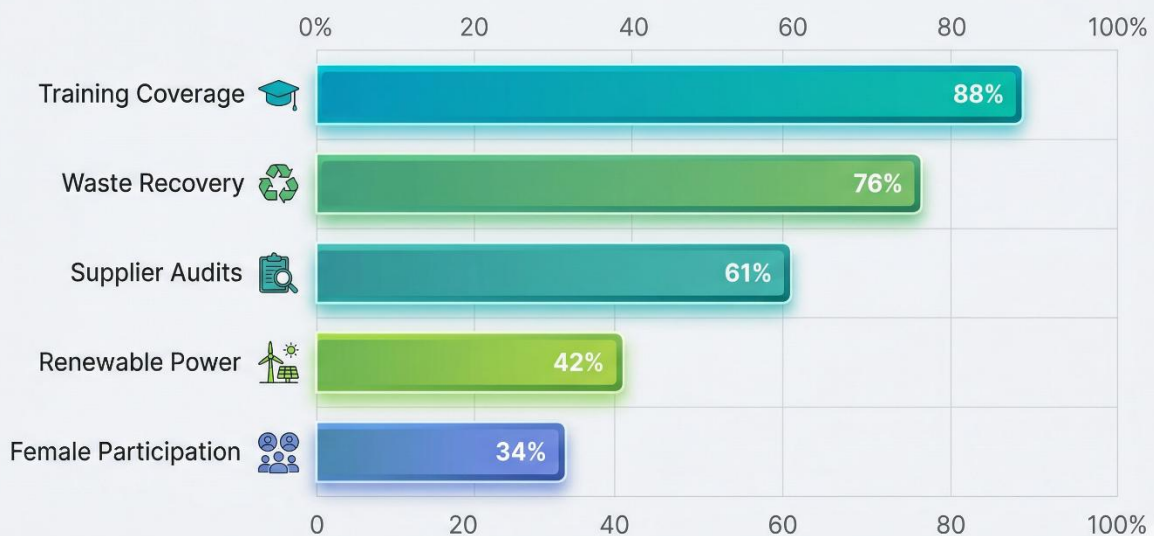
- **Demographics:** Heavy manufacturing exports (like metal and engineering) still skew heavily male, with industry averages showing women making up only **11.4% to 14% of the workforce**. Conversely, the apparel export sector is heavily female-dominated on the factory floor.
- **Training:** A standard benchmark for skill development in top-tier export firms is **6.5 to 8.5 training days per employee annually**.

6. Safety Incidents (LTIFR)

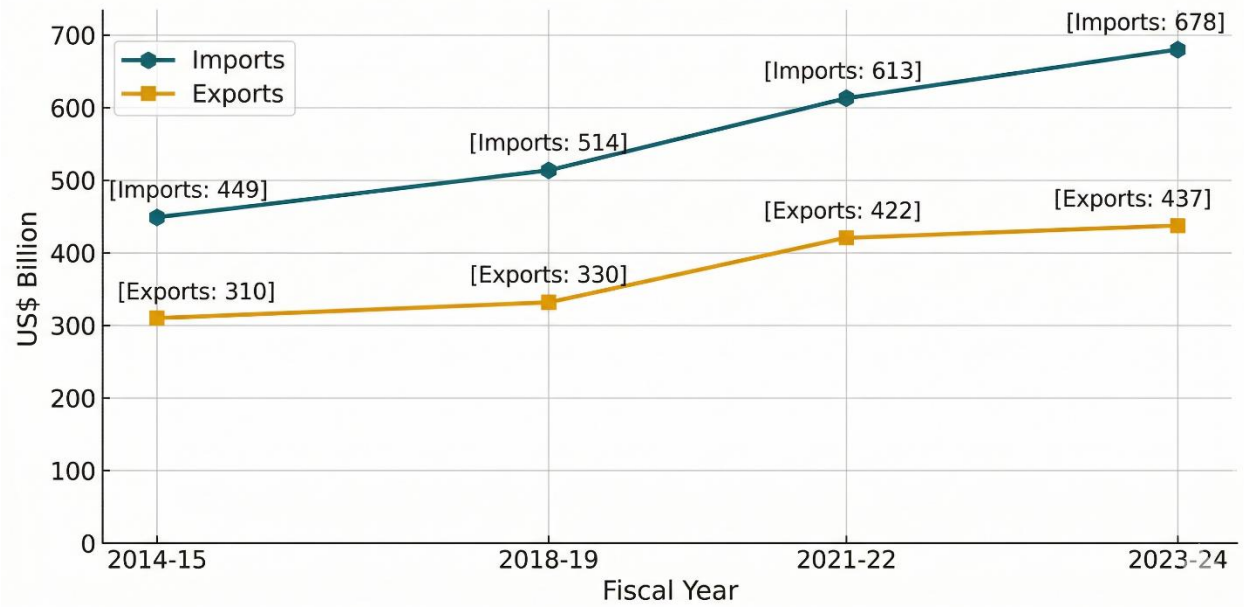
The **Lost Time Injury Frequency Rate (LTIFR)** is measured as the number of lost-time injuries per one million hours worked.

- **Indian Manufacturing Benchmarks:** Advanced Indian forging and engineering exporters report highly competitive LTIFR scores as low as **0.37**.
- **Global Averages:** The broader global heavy-manufacturing average sits around **0.72 to 0.85**. For an ESG report, an LTIFR target of **< 1.0** represents a highly controlled, safe working environment.

Key Performance Indicators - Progress to Goals



International Trade Data: Imports and Exports (US\$ Billion)



The Scoring Methodology:

1. The 0–10 Scoring Scale Each operational metric is scored from 0 to 10 based on performance against industry baselines and targets.

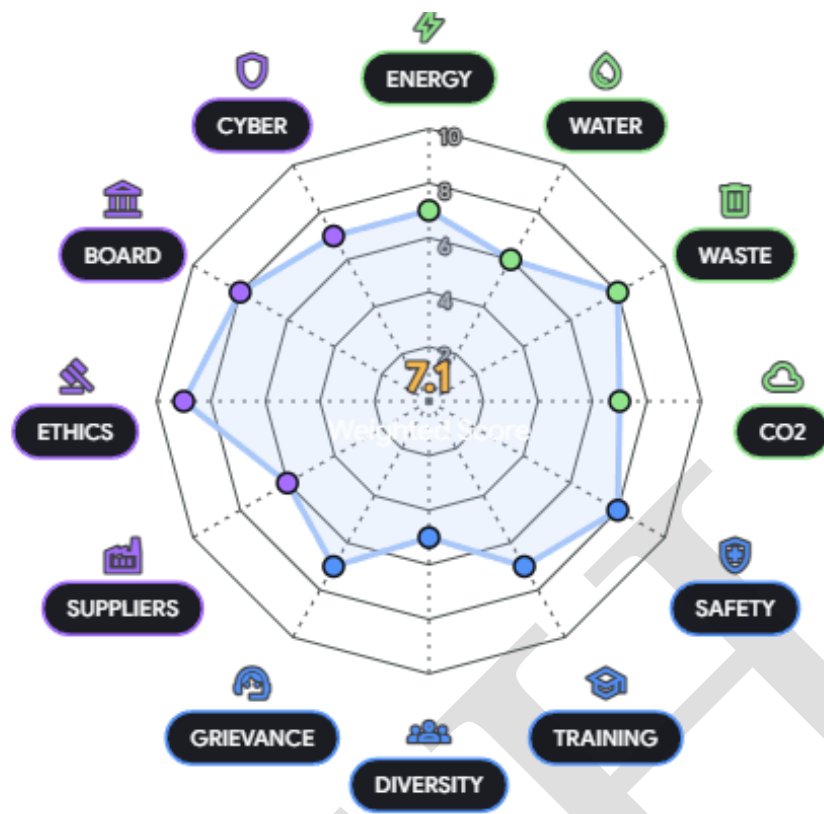
- **10:** Best-in-class / Target exceeded (e.g., 100% renewable energy, zero safety incidents).
- **5:** Industry average / Baseline met.
- **0:** Non-compliant / Severe risk (e.g., major regulatory fines, zero waste recovered).

2. The Weighting Strategy In heavy manufacturing and export industries, environmental and supply chain impacts are heavily scrutinized by overseas buyers. Therefore, the weights are distributed as follows:

- **Environmental (40%):** Energy, Water, Waste, Emissions.
- **Social (30%):** Safety (LTIFR), Training, Diversity, Grievances.
- **Governance (30%):** Supplier Audits, Compliance, Board Oversight, Data Security.

3. The Grading Matrix The final weighted score (out of 10) determines the company's ESG grade:

- **Grade A (8.5 – 10.0):** Export Leader (Highly attractive to premium global buyers).
- **Grade B (6.0 – 8.4):** Developing / Average (Meets standard compliance, room for operational efficiency).
- **Grade C (< 6.0):** High Risk / Laggard (Risk of losing export contracts or facing regulatory penalties).



Overall ESG Score Card

Inference:

Impact on Supply Chains

In an export business, ESG issues do not stay within one single department; they directly affect procurement, operations, and logistics.

- **The Shift to Deep Traceability:** Traceability is no longer optional; it is a core expectation in export markets, particularly for textiles, food, electronics, and minerals. Supply chains will have to adopt systems where origin, quality, labor, and environmental data move alongside the physical goods.
- **Supplier Consolidation and Audits:** Exporters can no longer plead ignorance about their suppliers' practices. Supplier audits must now actively cover labor standards, environmental controls, and business ethics. This will likely force exporters to consolidate their vendor base, dropping non-compliant suppliers in favor of those who can provide clean, audit-ready data.
- **Logistics and Multimodal Scrutiny:** Freight reliability and lead times are massive non-price competitiveness factors. However, variables like container availability, inland transport, and port congestion must now be managed not just for speed, but because they directly influence the company's reported emissions and service quality.

2. Impact on the Revenue Model

Because the export sector operates at such a massive scale across multiple sites, even a small change in ESG performance can materially move costs and financial exposure.

- **Accessing Premium Pricing (The Growth Lever):** ESG is not just a compliance cost; it is a direct market opportunity. Adopting green packaging, energy-efficient manufacturing, and certified sourcing strengthens access to premium, high-paying global buyers. Exporters who can prove their ESG metrics will win contracts over competitors who cannot.
- **Capital Allocation and Transition Costs:** The revenue model will have to absorb the costs of "transition risk." This happens when global buyers, financiers, and regulators begin demanding lower-carbon products and routes. Meeting these demands requires changes in capital allocation to fund high-efficiency machinery, renewable power, and advanced data analytics.
- **Protecting Margins from Disruption:** Physical climate risks (floods, heat, cyclones, port disruptions) directly interrupt export schedules. Building disaster resilience into the revenue model is necessary to protect delivery performance and retain customers.

Furthermore, failing to meet compliance standards can result in severe financial penalties, lost rebates, or revoked customs licenses.

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